Gr Fairfax Science Olympiad

Astronomy Key

CSE 2025

Name: _____ Score: 116/116

MCQ

1) Who created the Catalog of Nebulae and Star Clusters? (2pt)

    a) Charles Messier

    b) John Louis Emil Dreyer

    c) Johannes Kepler

    d) Fritz Zwicky

2) What is an AU? (1pt)

    a) The distance between the Earth and the Moon

    b) The distance between the Earth and the Sun

    c) The distance between the Sun and the closest star, Proxima Centauri

    d) The distance between the Earth and the edge of the observable universe

3) True or False: Chromatic aberrations are unique to refracting telescopes. (2pt)

    a) True

    b) False

4) What does superadiabaticity on the sun lead to? (2pt)

    a) Suppression of convection and stable layers

    b) Intense horizontal winds and jet streams

    c) Strong convection currents that transport energy outward

    d) Nuclear fusion near the solar surface

5) What wavelengths are protostars detected in? (1pt)

   a) Optical

   b) Ultraviolet

   c) Infrared

   d) None of the above

6) What masses of stars use the proton-proton chain? (1pt)

   a) Mass is not a factor

   b) <1.5 M⊙

   c) >1.5 M⊙

   d) <3.08 M⊙

7) Which spectral classes make up the Cepheid variables? (1pt)

   a) O - A

   b) B - F

   c) A - G

   d) F - K

   e) G - M

8) What happens if the Jeans criterion is met? (2pt)

   a) A black hole will form

   b) The helium flash will occur

   c) A protostar will form

   d) A supernova will occur

9) WC stars typically show strong emission lines of what elements? (2pt)

   a) He and N

   b) H and He

   c) Ca and O

   d) He and C

10) Where are variable stars found on the HR diagram? (1pt)

   a) Asymptotic Giant Branch

   b) Instability Strip

   c) Horizontal Branch

   d) Main Sequence

11) What is used to trace molecular clouds? (2pt)

   a) H_2

   b) N_2

   c) CO_2

   d) CO

12) Which element do T Tauri stars burn? (1pt)

   a) H

   b) He

   c) Li

   d) Ca

13) In the Yerkes classification, what does II represent? (2pt)

a) Hypergiants

b) Supergiants

c) Bright Giants

d) Regular Giants

14) Which of the following are used as standard candles? (1pt)

a) Red Giant

b) Black hole

c) RR Lyrae

d) Bok globule

15) Select all that apply to neutron stars: (2pt)

a) They are made mostly of tightly packed electrons

b) They have strong magnetic fields

c) They are larger than the Sun in diameter

d) They can emit beams of radiation detected as pulsars

16) What does a bright red emission line represent? (2pt)

a) O

b) N

c) H

d) He

17) Why do planetary nebulae appear green? (2pt)

a) They reflect green light not seen by stars

b) Green light is emitted from doubly ionized oxygen

c) Rayleigh scattering of light

d) Green light is emitted by carbon monoxide

18) Why does the helium flash occur? (2pt)

a) The star's core is rapidly rotating, causing turbulence

b) The core is electron-degenerate, so pressure does not increase with temperature

c) The outer hydrogen shell collapses, releasing gravitational energy suddenly

d) None of the above

19) True or False: Type 1a supernovae can form from the core collapse of a supergiant. (1pt)

a) True

b) False

20) In Type IIb supernovae, which is dominant, hydrogen or helium? (2pt)

a) Hydrogen

b) Helium

c) Neither; they are equally present

d) Neither; both are not present

21) What is driven by the kappa opacity mechanism? (2pt)

a) Sunspot cycle

b) Pulsations in some variable stars

c) Generation of magnetic fields

d) Lowered visibility of protostars

22) In which objects does the triple alpha process occur? (1pt)

a) Cepheids with extremely high luminosities

b) Red Giants with helium cores

c) Protostars with large disks

d) Binary white dwarves

23) What is the main reason for limb darkening? (1pt)

a) The outer layers of a star are cooler and emit less light than the inner layers, and we view them at a shallow angle at the limb.

b) The moon blocks the outer part of the Sun

c) The Sun has fewer fusion reactions occurring near the surface.

d) Atmospheric scattering makes the center of the Sun appear brighter from Earth.

24) True or False: Silicon is the last element created in stars. (1pt)

a) True

b) False

25) What is deuterium? (1pt)

a) An isotope of hydrogen with one proton and one neutron

b) Another name for H_2

c) A rare form of helium with two neutrons

d) A hydrogen molecule with an extra electron

FRQ

1) What is the difference between DA and DB and DC white dwarfs? (4pt)

   DA white dwarfs mainly exhibit hydrogen lines, DB white dwarfs show helium lines, and DC white dwarfs have a continuous spectrum

2) What is the Kelvin Helmholtz timescale? (3pt)

   The amount of time it would take a star to radiate away all of its gravitational potential energy at its current luminosity (if no energy is generated by fusion)

3) Explain the differences between the different types of T Tauri stars. (3pt)

   Classical T Tauri stars actively accrete material and show strong emission lines, and Weak-line T Tauri stars have little or no accretion and weak or no emission lines.

4) Why do lower mass pre-main-sequence stars follow the Hayashi track? (3pt)

Lower-mass pre-main-sequence stars follow the Hayashi track because they are fully convective, causing them to contract at nearly constant temperature while decreasing in luminosity as they approach the main sequence

5) What was the ultraviolet catastrophe? (4pt)

   An issue with the Rayleigh-Jeans law, it said that an idealized blackbody would emit infinite energy at ultraviolet wavelengths. Later fixed by Planck's introduction of quantized energy levels.

6) Why are radio telescopes so large? (3pt)

   The resolution of a telescope is proportional to the wavelength/diameter of the dish, so radio telescopes need to be larger to pick them up.

7) What causes distant stars to appear redder? (3pt)

Interstellar reddening, which happens when ISM scatters and absorbs more blue light than red light, making the star's light look redder than it actually is.

8) What happens in roche lobe overflow? (3pt)

A star gets so big that it overflows the Roche lobe, causing mass to be transferred onto the companion star

9) What happens when an object exceeds the Eddington limit? (4pt)

It loses mass because the pressure of radiation is stronger than gravity.

10) Star Alpha and Star Skibidi form a binary system and orbit around their common center of mass in nearly circular orbits. The total separation between the stars is 20 AU. Their orbital period is observed to be 40 years. If Alpha has a mass of 1.5 solar masses, find: (9pt)
    a) The mass of Star B in solar masses. **3.5 Solar masses**
    b) The distance of each star from the center of mass. **Alpha: 14 AU Skibidi: 6 AU**
    c) Based on the masses of each star, what are their respective Eddington limits? **Alpha: 4.93×10^4 $L_\odot$ Skibidi: 1.15×10^5 $L_\odot$**

11) Former CEO Andy Byron observes a star from Earth twice, six months apart. He finds that the star appears to shift its position relative to more distant background stars by an angle of 0.4 arcseconds. (6pt)
    a) What is the parallax angle in arcseconds? **0.2**
    b) Use the parallax angle to calculate the distance to the star in parsecs. **5**

12) A distant star, Omega, emits most of its light at a wavelength of 480 nanometers. Assume Omega behaves like an ideal blackbody. (6pt)

    a) Calculate the surface temperature of the star in Kelvin. **6040 K**

    b) Based on the peak wavelength, what color would this star appear to the human eye? **White blue**

13) The star Esnupi has a surface temperature of 6500 K and a radius equal to the Sun's radius. (6pt)

    a) Calculate the total luminosity of the star. **6.13×10^{26} W**

    b) If another star has twice the radius but the same temperature, how many times more luminous is it? **4**

14) A star, Bluey, is found to have a luminosity of 8.4×10^{24} W, and at a distance $7.1 * 10^{13}$ m away. (6pt)

    a) Calculate the apparent brightness in W/m^2. **1.33×10^{-4} W/m^2**

    b) What is the difference between luminosity and apparent brightness? Luminosity is the intrinsic measurement, whereas apparent brightness varies depending on distance.

15) Mini Gargantua is a black hole discovered in our universe, with a mass of (only!) 100 solar masses. (6pt)

    a) Calculate the Schwarzschild Radius of Mini Gargantua. **295km**

    b) What does the Schwarzschild radius tell us about a black hole? **The Schwarzschild radius tells us the radius of the event horizon for a non rotating black hole. Within this distance, nothing can escape from the black hole.**

16) An astronaut is stranded on the exoplanet Edmund, which has a radius of 1.4 Earth radii, and a mass 2 times that of Earth's. (9pt)

a) Calculate the escape velocity the astronaut needs for a 1200kg ship. **13.4 km/s**

b) How, if at all, does the mass of the ship affect the escape velocity? **The mass does not affect the escape velocity, only the mass of the exoplanet does.**

c) Calculate the difference between escape and orbital velocity? 4 km/s